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2 Generalized Local Polynomial Regression with Decomposed 3 Context-Aware Kernels

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6 Abstract

Local Polynomial Regression (LPR) is a powerful tool for nonparametric smoothing, yet it traditionally suffers from a "Euclidean tautology": the variables used to define the local neighborhood are identical to those used in the polynomial fit. This restricts its ability to handle complex domains where the regression function varies across non-Euclidean structures, such as graphs, manifolds, or discrete categories, while remaining locally smooth in the primary feature space. We propose *Generalized Context-Aware LPR* (GC-LPR), a framework that decouples the fitting coordinates (Z) from the weighting context (C). By adopting a modeling convention where the conditional mean depends jointly on Z and C ($Y = m_C(Z) + \varepsilon$), our estimator acts as a "projected smoother": it isolates a slice of the data on the manifold defined by C via a compound product kernel, and performs polynomial fitting in the Z -coordinates within that slice. This enables practitioners to model responses that vary across graphs, networks, or categorical strata while retaining the interpretability and bias properties of LPR in a primary Euclidean feature space. Theoretical analysis clarifies the induced context-smoothed target of GC-LPR and shows that the method preserves the Euclidean bias-reduction properties of standard LPR while allowing arbitrary, non-Euclidean contexts to modulate the local estimation. We demonstrate the efficacy of this approach on geospatial and network-structured datasets.

1. Introduction

Local Polynomial Regression (LPR) is a cornerstone of nonparametric smoothing, prized for its ability to adapt locally without committing to a rigid global functional form. Its theoretical virtues are well-established, especially for local linear and higher-order variants: the local linear estimator is design-adaptive and automatically corrects first-order boundary bias, while the broader local-polynomial family attains the usual minimax-optimal rates under standard smoothness conditions (Fan and Gijbels, 1996; Loader, 1999). However, classical LPR suffers from a restrictive “Euclidean tautology”: the variables used to define the local neighborhood are identical to those used in the polynomial fit.

This constraint is increasingly inadequate in modern domains where the notion of proximity is multi-faceted. Observations may be “near” geographically, connected via a transport network, or similar in terms of categorical attributes, even if they are distant in the feature space of interest. Furthermore, relying on a monolithic kernel over all available features exacerbates the “curse of dimensionality” (Stone, 1982), degrading performance as the dimension of the predictor space grows. Generalized product kernels have been successfully applied to mixed-data regression (Racine and Li, 2004), and GWR variants have already explored non-Euclidean notions of locality by replacing Euclidean distance with road-network distance, travel time, metro-network distance, or network-derived weighting schemes (Lu et al., 2011, 2014; Gao et al., 2022; He et al., 2023).

Our contribution is therefore more specific. We introduce *Generalized Context-Aware LPR* (GC-LPR), a decomposed local-polynomial framework in which the polynomial fit is performed in primary Euclidean coordinates Z , while one or more auxiliary contexts $C_1, \dots, C_J$ enter through a compound product kernel K_c (formal definition in Section 3). The Euclidean factor governs local polynomial fitting in Z , and the context factors restrict the neighborhood using domain-appropriate similarity notions, ranging from discrete labels to nodes on a graph. This yields an induced context-smoothed target m_W and a bias-variance interpretation that separates polynomial approximation in Z from smoothing across C . Accordingly, the asymptotic theory in Section 5 is stated for m_W under fixed context bandwidths rather than for the point target $m_{c^*}(z)$.

Conceptually, we do not treat context as affecting only the variance. Instead, we model the regression function itself as context-dependent, writing $m_C(Z)$. The context kernels then isolate a relevant slice of the data defined by C , within which the local polynomial fit is carried out in Z . This allows the relationship between Z and Y to change across the context space, whether smoothly or through structural discontinuities such as network or geographic boundaries. In this sense, GC-LPR is distinct from global graph-regularization methods such as Graph Signal Processing (GSP) (Shuman et al., 2013).

We further demonstrate the composability of this framework by integrating a response-based robustness factor K_r , following recent work by Shulman (2025), to handle outliers without distorting the definition of locality. The remainder of this paper reviews the relevant literature, formalizes the estimator, analyzes its asymptotic properties, and validates its performance on geospatial and network-structured datasets.

2. Background and Notation

This section establishes the formal notation and background for the methods discussed. We first review the canonical framework for Local Polynomial Regression (LPR) to define its ob-

jective and properties. Finally, we summarize the RSKLPR framework, which introduces a response-based kernel for robustness.

2.1. Local Polynomial Regression

Let $\mathcal{D}_n = \{(Z_i, Y_i)\}_{i=1}^n$ be i.i.d. with $Z_i \in \mathbb{R}^d$ and $Y_i \in \mathbb{R}$. Fix $z \in \mathbb{R}^d$ and write $t_i := Z_i - z$. Let $\mathcal{A}_p = \{\alpha \in \mathbb{N}_0^d : |\alpha| \leq p\}$, $t^\alpha = \prod_{j=1}^d t_j^{\alpha_j}$, and

$$r_p(t) := (t^\alpha)_{\alpha \in \mathcal{A}_p} \in \mathbb{R}^M, \quad M = \binom{p+d}{d}.$$

The local polynomial is $g_z(u; \beta) = r_p(u - z)^\top \beta(z)$, and the estimator of $m(z)$ is $\hat{m}(z) := \hat{\beta}_0(z)$, the intercept.

Let $K : \mathbb{R}^d \rightarrow [0, \infty)$ be a kernel and H a positive-definite bandwidth matrix. With $K_H(t) := |H|^{-1} K(H^{-1}t)$, where $|H| := \det(H)$ and in the isotropic case $H = hI_d$ gives $K_H(t) = h^{-d} K(t/h)$. In empirical objectives we use the unnormalized weight $K(H^{-1}t)$, since $|H|$ is constant in i , so it does not affect the minimizer. The empirical LPR objective is:

$$\mathcal{L}_{lpr}(z; \mathcal{D}_n, H) = \sum_{i=1}^n (Y_i - r_p(t_i)^\top \beta(z))^2 K(H^{-1}t_i). \quad (1)$$

Let R be the $n \times M$ design with i th row $r_p(t_i)^\top$, $W_z := \text{diag}\{K(H^{-1}t_1), \dots, K(H^{-1}t_n)\}$, and $Y := (Y_1, \dots, Y_n)^\top$. When $R^\top W_z R$ is nonsingular,

$$\hat{\beta}(z) = (R^\top W_z R)^{-1} R^\top W_z Y, \quad \hat{m}(z) = e_1^\top \hat{\beta}(z).$$

The population analogue (with the normalized kernel) is

$$\mathcal{J}_{std}(z; \beta) = \iint (v - g_z(u; \beta))^2 K_H(u - z) f_{ZY}(u, v) dv du. \quad (2)$$

Under standard conditions (continuity of f_Z with $f_Z(z) > 0$; $m \in C^{p+1}$ near z ; symmetric integrable K with unit mass; $H \rightarrow 0$ and $n|H| \rightarrow \infty$ in the isotropic case), the local linear estimator ($p = 1$) is design-adaptive and, for interior z ,

$$\text{Bias}\{\hat{m}(z)\} = O(\|H\|^2), \quad \text{Var}\{\hat{m}(z)\} = O((n|H|)^{-1}).$$

See (Fan, 1993; Fan and Gijbels, 1996; Loader, 1999) for details. Product and radial kernels both fit within this framework.

2.2. Robust Local Polynomial Regression with Similarity Kernels

Robust Local Polynomial Regression with Similarity Kernels (RSKLPR) (Shulman, 2025) augments LPR with a similarity kernel on the full data space to achieve robustness, in the spirit of classical robust local regression and local polynomial M-estimation (Cleveland, 1979; Jiang and Mack, 2001). Writing its empirical loss at (z, y) as

$$\mathcal{L}_{rsk}(z, y; \mathcal{D}_n, \mathcal{H}) = \sum_{i=1}^n \left(Y_i - r_p(t_i)^\top \beta(z, y) \right)^2 \mathcal{K}_D((z, y), (Z_i, Y_i); \mathcal{H}), \quad (3)$$

74 with the decomposition

$$\mathcal{K}_D((z, y), (u, v); \mathcal{H}_1, \mathcal{H}_2) = K_1(z - u; \mathcal{H}_1) K_2((z, y), (u, v); \mathcal{H}_2), \quad (4)$$

75 a separable choice $K_2((z, y), (u, v)) = \hat{f}_{Y|Z}(y | z) \hat{f}_{Y|Z}(v | u)$ yields, after dropping the multiplicative
76 constant $\hat{f}_{Y|Z}(y | z)$ (independent of i), a simplified empirical loss independent of y at the query:

$$\tilde{\mathcal{L}}(z) = \sum_{i=1}^n \left(Y_i - r_p(t_i)^\top \beta(z) \right)^2 K_1(t_i; \mathcal{H}_1) \hat{f}_{Y|Z}(Y_i | Z_i; \mathcal{H}_2). \quad (5)$$

77 In the population, the weight appears squared because it is multiplied by the conditional density
78 itself as a weight inside the expectation:

$$\mathcal{J}_{rsk}(z; \beta) = \iint (v - g_z(u; \beta))^2 K_1(z - u; \mathcal{H}_1) [f_{Y|Z}(v | u)]^2 f_Z(u) dv du. \quad (6)$$

79 The intercept targets $\mu'(z) = \frac{\int v [f_{Y|Z}(v|z)]^2 dv}{\int [f_{Y|Z}(v|z)]^2 dv}$, which reduces to $m(z)$ when $f_{Y|Z}(\cdot | z)$ is symmetric
80 about its mean; see (Shulman, 2025).
81

82 3. Generalized Context-Aware LPR

83 3.1. Modeling Convention: The Context-Dependent Mean

84 Let $X \in \mathcal{X}$ denote a predictor observation where we emphasize that $\mathcal{X}$ is a general predictor
85 space: an observation X may include Euclidean features, categorical labels, spatial coordinates,
86 or identifiers of nodes in a graph or network. We assume there exists a Euclidean coordinate
87 chart $\psi_0 : \mathcal{X} \rightarrow \mathbb{R}^d$ and additional feature maps yielding the decomposition $X \mapsto (Z, C)$, where
88 $Z = \psi_0(X) \in \mathbb{R}^d$ are the fitting coordinates and C collects additional *contexts* (categorical,
89 geospatial, network-valued, etc.). We do not treat C merely as variance modifiers but rather
90 adopt the conditional-mean formulation:

$$m_c(z) \equiv \mathbb{E}[Y | Z = z, C = c].$$

91 Accordingly, we write the observation model as

$$Y = m_C(Z) + \varepsilon, \quad \mathbb{E}[\varepsilon | Z, C] = 0, \quad \text{Var}(\varepsilon | Z, C) = \sigma^2(Z, C).$$

92 The underlying point target of interest at $x^* = (z, c^*)$ is the context-specific regression function:

$$m_{c^*}(z).$$

93 Here, the regression function $m_{c^*}(z)$ is assumed to be smooth (locally polynomial) in Z , but
94 its dependence on C may be complex, discontinuous, or defined on a graph. The role of the
95 context kernels is to isolate a local "slice" or neighborhood in the C -space, within which the
96 polynomial approximation in Z is valid.

3.2. Estimator

We decouple the Euclidean fitting chart from the weighting context by defining feature maps over the general predictor space $\mathcal{X}$. We define the *compound predictor kernel* as a product of individual kernels on mapped features:

$$K_c(x, u; \mathcal{H}_c) = K_{0,H}(\psi_0(x) - \psi_0(u)) \prod_{j=1}^J K(\psi_j(x), \psi_j(u); \eta_j), \quad \mathcal{H}_c = (H, \eta_1, \dots, \eta_J). \quad (7)$$

Here, $\psi_0 : \mathcal{X} \rightarrow \mathbb{R}^d$ denotes the Euclidean coordinate chart used for local polynomial fitting, and we write $Z = \psi_0(X)$ for the corresponding chart coordinates, equipped with the standard kernel $K_{0,H}(\cdot) = |H|^{-1} K_0(H^{-1}\cdot)$. For $j \geq 1$, the maps $\psi_j : \mathcal{X} \rightarrow C_j$ extract auxiliary context features $C_j \in \mathcal{C}_j$. The context kernels $K_j(\cdot; \eta_j)$ are symmetric on their respective spaces C_j and operate on the feature maps $\psi_j(X)$; they may incorporate non-Euclidean structure (e.g., Haversine distance for coordinates, graph shortest-path or diffusion distances (Shuman et al., 2013), or Aitchison–Aitken kernels for discrete labels (Aitchison and Aitken, 1976)). This recovers classical product-kernel mixed-data smoothing while generalizing to non-Euclidean topologies.

With these components, the GC-LPR estimator $\hat{\beta}(x)$ minimizes the context-weighted least squares objective:

$$\mathcal{L}_{\text{gc-lpr}}(x) = \sum_{i=1}^n \left(Y_i - r_p(\psi_0(X_i) - \psi_0(x))^\top \beta(x) \right)^2 K_c(x, X_i; \mathcal{H}_c). \quad (8)$$

Equivalently, writing $z = \psi_0(x)$ and $Z_i = \psi_0(X_i)$, define the empirical local Gram matrix

$$G_n(x) := \sum_{i=1}^n K_c(x, X_i; \mathcal{H}_c) r_p(Z_i - z) r_p(Z_i - z)^\top.$$

When $G_n(x)$ is nonsingular, the minimizer is unique and satisfies

$$\hat{\beta}(x) = G_n(x)^{-1} \sum_{i=1}^n K_c(x, X_i; \mathcal{H}_c) r_p(Z_i - z) Y_i.$$

In computation one may replace $G_n(x)^{-1}$ by a Moore–Penrose inverse if needed; for the pointwise asymptotic results below, we assume $G_n(x^*)$ is nonsingular with probability tending to one. Crucially, the polynomial basis $r_p(\cdot)$ operates only on $Z_i = \psi_0(X_i)$, whereas the weight factorizes as

$$K_c(x, X_i; \mathcal{H}_c) = K_{0,H}(Z_i - z) W(x, X_i), \quad W(x, u) := \prod_{j=1}^J K(\psi_j(x), \psi_j(u); \eta_j),$$

where $z = \psi_0(x)$ and W involves only the maps $\{\psi_j\}_{j \geq 1}$. Note that while the polynomial basis r_p varies only with Z , the estimated coefficients $\beta(x)$ vary with the full context C , allowing the shape of the regression function to change across contexts.

Specifically, given finite context bandwidths, the estimator naturally targets the *context-smoothed mean*:

$$m_W(z; x^*) = \frac{\mathbb{E}[Y W(x^*, X) \mid Z = z]}{\mathbb{E}[W(x^*, X) \mid Z = z]},$$

where $x^* = (z, c^*)$ with $z = \psi_0(x^*)$. While analytically distinct from the pointwise target $m_{c^*}(z)$, in domains such as graph signal processing this smoothing is often a desired feature rather than a bias term. It allows the model to “borrow strength” from topologically proximate observations, effectively denoising the structural dependency on C while fitting the geometry of Z .

Hence, at x^* , GC-LPR yields a local polynomial approximation to $m_W(z; x^*)$; this coincides with $m_{c^*}(z)$ under exact context selection or vanishing context bandwidth.

Example (geospatial and categorical contexts). Consider a raw observation X consisting of tabular features Z_{feat} , geospatial coordinates Z_{geo} , a region label C_{cat} , and a network node C_{net} . We define the feature maps as:

$$\psi_0(X) = Z_{\text{feat}} \oplus Z_{\text{geo}}, \quad \psi_1(X) = Z_{\text{geo}}, \quad \psi_2(X) = C_{\text{cat}}, \quad \psi_3(X) = C_{\text{net}},$$

where $\oplus$ denotes concatenation. We then define the compound kernel K_c explicitly in terms of these inputs:

$$K_c(x, u; \mathcal{H}_c) = K_{0,H}(\psi_0(x) - \psi_0(u)) \cdot K_{\text{geo}}(\psi_1(x), \psi_1(u); \eta_1) \cdot K_{\text{cat}}(\psi_2(x), \psi_2(u); \eta_2) \cdot K_{\text{net}}(\psi_3(x), \psi_3(u); \eta_3).$$

This formulation allows the model to handle “manifold folding” (e.g., cul-de-sacs) where points are close in the Euclidean coordinates induced by ψ_0 but distant in the network context ψ_3 . Specifically, K_{net} will assign low weight to physically close but network-distant points, correctly estimating the mean for the specific network context. Note that Z_{geo} appears in both the fitting chart ψ_0 and the context map ψ_1 ; GC-LPR allows variables to be shared between the regressor r_p and the weighting kernel K_c . When overlaps are heavy, one may temper factors via exponents $K_j^{\alpha_j}$ with $\alpha_j \in (0, 1]$, or tune the (H, η_j) by cross-validation.

3.3. A Robust Extension (GRC-LPR)

Optional robustness against Y -outliers is incorporated by a separable response-based factor K_r , following Shulman (2025). To control the curse of dimensionality in the density estimation, we define a robustness feature map $\psi_r : \mathcal{X} \rightarrow \mathbb{R}^{d_r}$ (typically $\psi_r(X) = \psi_0(X) = Z$) to select the variables relevant for detecting outliers. Define

$$\mathcal{K}_{\mathcal{D}}^*((x, y), (u, v); \mathcal{H}) := K_c(x, u; \mathcal{H}_c) K_r((\psi_r(x), y), (\psi_r(u), v); \mathcal{H}_r), \quad (9)$$

and consider the empirical loss at x obtained when K_r is chosen as a separable conditional-density weight in the ψ_r -chart, i.e., $K_r((\psi_r(x), y), (\psi_r(u), v)) = \hat{f}_{Y|\psi_r(X)}(y | \psi_r(x); \mathcal{H}_r) \hat{f}_{Y|\psi_r(X)}(v | \psi_r(u); \mathcal{H}_r)$. Dropping the multiplicative constant $\hat{f}_{Y|\psi_r(X)}(y | \psi_r(x); \mathcal{H}_r)$ (independent of i) yields:

$$\tilde{\mathcal{L}}_{\text{grc-lpr}}(x) = \sum_{i=1}^n \left(Y_i - r_p(\psi_0(X_i) - \psi_0(x))^\top \beta(x) \right)^2 K_c(x, X_i; \mathcal{H}_c) \hat{f}_{Y|\psi_r(X)}(Y_i | \psi_r(X_i); \mathcal{H}_r). \quad (10)$$

This formulation yields a single-step reweighted LPR where points with low conditional density, relative to the chosen subspace $\psi_r(X)$, receive smaller weights. While one may choose $\psi_r(X) = X$, restricting ψ_r to the fitting coordinates Z or a low-dimensional subset may be preferred to ensure stable density estimates.

4. Related Work

The proposed GC-LPR framework sits at the intersection of nonparametric smoothing, mixed-data regression, and signal processing on irregular domains. This section contextualizes our contribution within these established fields.

156 4.1. Generalized Product Kernels and Mixed Data

157 The foundational challenge in nonparametric regression is the “curse of dimensionality.”
158 [Racine and Li \(2004\)](#) revolutionized the treatment of mixed data by introducing generalized
159 product kernels. Their estimator, $\hat{E}(Y|X^c, X^d)$, utilizes standard kernels for continuous variables
160 (X^c) and variation-based kernels (e.g., Aitchison-Aitken) for discrete variables (X^d). A critical
161 property of this framework is “automatic relevance determination”: cross-validation can effectively
162 shrink the bandwidths of irrelevant predictors to infinity, smoothing them out of the model
163 ([Hall et al., 2007](#)).

164 However, the Racine-Li framework assumes that all variables entering the kernel also enter
165 the regression function. GC-LPR relaxes this by decoupling the sets. We allow context variables
166 C to enter the kernel K_c without forcing them into the polynomial basis $r_p(Z)$. This is broadly
167 in line with work on associated kernels for mixed or support-constrained regressors ([Somé and](#)
168 [Kokonendji, 2016](#)), which emphasizes matching the kernel support to the support of each regres-
169 sor. In GC-LPR, however, the context kernels serve as auxiliary locality weights rather than as
170 additional regressors entering the polynomial mean directly.

171 4.2. Varying Coefficient Models and GWR

172 Varying Coefficient Models (VCM) extend the linear model by allowing coefficients to change
173 smoothly as a function of an index variable, typically written as $Y = X^T\beta(U) + \epsilon$ ([Hastie and](#)
174 [Tibshirani, 1993](#); [Fan and Zhang, 1999](#)). [He and Huang \(2009\)](#) further refined VCM estimation
175 using double-smoothing bias reduction techniques.

176 Geographically Weighted Regression (GWR) ([Brunsdon et al., 1996](#); [Fotheringham et al.,](#)
177 [2003](#)) is perhaps the most famous instance of a spatial VCM. In classical GWR and related
178 extensions such as Multiscale GWR (MGWR), one local regression is indexed by location and
179 locality is defined through a spatial weighting scheme ([Fotheringham et al., 2017](#)). Prior GWR
180 work has already broadened this notion of locality beyond Euclidean distance, including road-
181 network distance, travel time, metro-network distance, and network-derived weight matrices ([Lu](#)
182 [et al., 2011, 2014](#); [Gao et al., 2022](#); [He et al., 2023](#)). Those contributions primarily modify the
183 distance metric or network weight matrix inside otherwise standard GWR machinery. A more
184 theoretical GWR literature also studies estimation and inference, location-specific bandwidths,
185 tests for locational heterogeneity and model specification, and local-linear bias reduction ([Páez](#)
186 [et al., 2002a,b](#); [Wang et al., 2008](#)).

187 GC-LPR is related to VCM and GWR, but it is not a direct re-labeling of the context C as
188 the VCM index. Standard VCMs and GWR typically use a single indexing space to define the
189 local fit. GC-LPR instead separates fitting coordinates Z from auxiliary contexts $C_1, \dots, C_J$, and
190 allows several context kernels to enter multiplicatively while the polynomial basis remains in Z .
191 In that sense, the framework is closer to a decomposed, context-indexed local polynomial model
192 than to a standard spatial VCM. Our novelty is therefore not the use of non-Euclidean locality
193 itself, but the decomposed multi-context local-polynomial formulation and the target-level theory
194 built around m_W .

195 4.3. Graph Signal Processing and Manifold Regularization

196 When data resides on irregular domains, Graph Signal Processing (GSP) and manifold regu-
197 larization address smoothness via operators such as the graph Laplacian ([Shuman et al., 2013](#)).
198 Diffusion kernels (or heat kernels) $K_t = e^{-tL}$ generalize Gaussian smoothing to manifolds and
199 graphs, capturing the intrinsic geometry of the data domain ([Kondor and Lafferty, 2002](#); [Belkin](#)

and Niyogi, 2003; Coifman and Lafon, 2006). These methods are conceptually distinct from local regression: they are typically global graph-smoothing or spectral procedures rather than weighted local polynomial fits.

Work by Bickel and Li (2007) extended local polynomial regression to unknown manifolds, proving that LPR can adapt to the intrinsic dimension of the data. GC-LPR is complementary to these spectral methods. By incorporating graph-based kernels (e.g., diffusion or shortest-path) as multiplicative factors in K_c , we combine local-polynomial fitting in Z with graph-informed context weighting. This allows us to estimate the context-dependent mean $m_C(Z)$ even when C represents a complex manifold structure that is not isometric to Euclidean space.

In summary, GC-LPR can be viewed as a decomposed local-polynomial framework related to mixed-data smoothing, GWR, and graph-based learning, but distinct in its separation of fitting coordinates from multiple auxiliary contexts and in its target-level interpretation via m_W .

5. Theoretical Properties

We study the estimator in (8) at a fixed query point $x^* \in \mathcal{X}$. Write $z = \psi_0(x^*) \in \mathbb{R}^d$, and let c^* denote the associated context value of x^* under the decomposition $X \mapsto (Z, C)$. Assume that z is an interior point of the support of Z . When discussing population quantities, we use the regular conditional distribution of (X, Y) given $Z = \psi_0(X)$, and write $\mathbb{E}[\cdot | Z = u]$ for conditional expectations. Throughout this section we assume i.i.d. sampling of $\{(X_i, Y_i)\}_{i=1}^n$. Consequently, the asymptotic results below do not cover graph-dependent or graph-temporal dependence structures such as those used in Experiments 3 and 4; those experiments should therefore be read as empirical demonstrations outside the scope of the present theory. Throughout this section, the context bandwidths $\eta_1, \dots, \eta_J$ are treated as fixed while the Euclidean bandwidth matrix H shrinks. Accordingly, the estimand of the asymptotic analysis is the induced context-smoothed target $m_W(\cdot; x^*)$; consistency for the point target $m_{c^*}(z)$ would require a separate theorem with $\eta_j \rightarrow 0$ and compatible joint rate conditions.

For clarity, recall the context weight and its conditional mean

$$W(x, u) := \prod_{j=1}^J K(\psi_j(x), \psi_j(u); \eta_j), \quad \gamma_x(u) := \mathbb{E}[W(x, X) | Z = u],$$

so that the compound kernel factorizes into Euclidean and context components:

$$K_c(x, u) = K_{0,H}(\psi_0(x) - \psi_0(u)) W(x, u).$$

Whenever $\gamma_x(u) > 0$ and the displayed conditional expectations are finite, define the associated context-smoothed conditional mean

$$m_W(u; x) := \frac{\mathbb{E}[Y W(x, X) | Z = u]}{\mathbb{E}[W(x, X) | Z = u]} = \frac{\mathbb{E}[Y W(x, X) | Z = u]}{\gamma_x(u)}.$$

Lemma 1 (Context-smoothed conditional target). *Fix $x \in \mathcal{X}$ and $u \in \mathbb{R}^d$ such that $0 < \gamma_x(u) < \infty$ and*

$$\mathbb{E}[Y^2 W(x, X) | Z = u] < \infty.$$

Then

$$m_W(u; x) = \arg \min_{\alpha \in \mathbb{R}} \mathbb{E}[(Y - \alpha)^2 W(x, X) | Z = u].$$

232 Equivalently, for every measurable function $a : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\mathbb{E}[(Y - a(Z))^2 W(x, X) \mid Z = u] = \mathbb{E}[(Y - m_W(u; x))^2 W(x, X) \mid Z = u] + \gamma_x(u) (a(u) - m_W(u; x))^2.$$

233 In particular, for each fixed u , $m_W(u; x)$ is the unique conditional weighted L^2 -projection onto
 234 constants; equivalently, $m_W(\cdot; x)$ is the pointwise conditional weighted L^2 -projection target in-
 235 duced by the context weights.

236 Proof in [Appendix A.2](#). Under the modeling convention $m_c(z) = \mathbb{E}[Y \mid Z = z, C = c]$, our
 237 estimand at $x^\star$ is the context-specific mean $m_{c^\star}(z)$. However, with finite context bandwidths, the
 238 induced population target is the context-smoothed mean $m_W(z; x^\star)$; the difference $m_W(z; x^\star) -$
 239 $m_{c^\star}(z)$ captures the effect of smoothing (or borrowing strength) across contexts.

240 5.1. Assumptions

241 **Assumption 1** (Data, design, and local smoothness). Let $\{(X_i, Y_i)\}_{i=1}^n$ be i.i.d. copies of (X, Y) .
 242 Let $Z = \psi_0(X) \in \mathbb{R}^d$ admit a density f_Z that is continuous in a neighborhood of z , with $f_Z(z) > 0$.
 243 Assume the regression model

$$Y = m_C(Z) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid Z, C] = 0, \quad \text{Var}(\varepsilon \mid Z, C) = \sigma^2(Z, C).$$

244 Fix $x^\star$, and assume the context-smoothed mean $m_W(\cdot; x^\star) \in C^{p+1}$ in a neighborhood of z . In
 245 particular, if the context kernels induce no leakage away from $c^\star$ so that $m_W(\cdot; x^\star) = m_{c^\star}(\cdot)$
 246 locally, this reduces to $m_{c^\star}(\cdot) \in C^{p+1}$ near z .

247 **Assumption 2** (Euclidean kernel and bandwidth). $K_0 : \mathbb{R}^d \rightarrow [0, \infty)$ is even, integrable with
 248 finite moments up to order $p + 2$, and $\int K_0(t) dt = 1$. The bandwidth H is positive definite with
 249 $\|H\| \rightarrow 0$ and $n\|H\| \rightarrow \infty$. Moreover, the kernel moment matrices

$$M_0 := \int r_p(t) r_p(t)^\top K_0(t) dt, \quad \Omega_0 := \int r_p(t) r_p(t)^\top K_0(t)^2 dt$$

250 have finite entries, and M_0 is nonsingular.

251 **Assumption 3** (Context factors and effective-weight regularity). For $j \geq 1$, the context factor
 252 $K_j(\cdot, \cdot; \eta_j)$ is symmetric and nonnegative, and is uniformly bounded above: $0 \leq K_j(\cdot, \cdot; \eta_j) \leq$
 253 $\bar{K}_j < \infty$. (Equivalently, one may replace each factor by $K_j/\bar{K}_j \in [0, 1]$, which only rescales the
 254 weighted least-squares objective by a constant and does not change its minimizer.)

255 Fix $x^\star \in \mathcal{X}$. Assume the conditional mean weight

$$\gamma_{x^\star}(u) = \mathbb{E}[W(x^\star, X) \mid Z = u]$$

256 is locally Lipschitz at z ; i.e., there exist constants $L < \infty$ and $\delta > 0$ such that

$$|\gamma_{x^\star}(u) - \gamma_{x^\star}(z)| \leq L\|u - z\| \quad \text{whenever } \|u - z\| < \delta.$$

257 This regularity requirement is imposed on the effective conditional mean weight $\gamma_{x^\star}$ itself, re-
 258 gardless of whether the underlying context variables are continuous, discrete, or mixed. Locally
 259 constant cases are included as a special case.

260 **Assumption 4** (Positivity). Let $\phi_1(x^\star) := \mathbb{E}[W(x^\star, X) \mid Z = z] = \gamma_{x^\star}(z)$. Assume $0 < \underline{\phi} \leq \phi_1(x^\star)$.

Combined with Assumptions 1 and 3, this pointwise positivity implies that, after possibly shrinking the neighborhood of z , both $\gamma_{x^*}(u)$ and the effective design density $q_{x^*}(u) := f_Z(u)\gamma_{x^*}(u)$ are bounded away from zero near z .

Assumption 5 (Effective residual second-moment regularity). Fix $x^* \in \mathcal{X}$, and define

$$v_{x^*}(u) := \mathbb{E}\left[W(x^*, X)^2 (Y - m_W(u; x^*))^2 \mid Z = u\right].$$

Assume v_{x^*} is locally bounded in a neighborhood of z and continuous at z . Equivalently, on the event $\{Z = u\}$, one may write

$$v_{x^*}(u) = \mathbb{E}\left[W(x^*, X)^2 (Y - m_W(Z; x^*))^2 \mid Z = u\right].$$

5.2. Bias and Variance

Lemma 2 (Effective population LPR problem). Fix a query point $x \in \mathcal{X}$ and write $z = \psi_0(x)$. The population objective

$$\mathcal{J}_{\text{gc-lpr}}(x; \beta) := \mathbb{E}\left[(Y - r_p(Z - \psi_0(x))^\top \beta(x))^2 K_{0,H}(Z - \psi_0(x)) W(x, X)\right]$$

is, up to an additive constant independent of β , equal to

$$\tilde{\mathcal{J}}_{\text{gc-lpr}}(x; \beta) = \int \left(m_W(u; x) - r_p(u - z)^\top \beta(x)\right)^2 K_{0,H}(u - z) \gamma_x(u) f_Z(u) du.$$

In particular, GC-LPR is (population-wise) standard local polynomial regression for the target $m_W(\cdot; x)$ under the effective design density $u \mapsto f_Z(u) \gamma_x(u)$.

Proof. Apply Lemma 1 with $a(Z) = r_p(Z - \psi_0(x))^\top \beta(x)$, multiply by $K_{0,H}(Z - \psi_0(x))$, and take expectations. The additive term

$$\mathbb{E}\left[(Y - m_W(Z; x))^2 K_{0,H}(Z - \psi_0(x)) W(x, X)\right]$$

does not depend on β , and the remaining term reduces by iterated expectation to

$$\int \left(m_W(u; x) - r_p(u - z)^\top \beta(x)\right)^2 K_{0,H}(u - z) \gamma_x(u) f_Z(u) du.$$

□

Proposition 1 (Bias decomposition and rate). Under Assumptions 1–4, fix $x^* \in \mathcal{X}$. The GC-LPR intercept estimator $\hat{m}(x^*) := e_1^\top \hat{\beta}(x^*)$, where $e_1 = (1, 0, \dots, 0)$ selects the intercept term, has induced population target $m_W(z; x^*)$. Moreover, its bias relative to the context-specific mean $m_{c^*}(z)$ decomposes as

$$\mathbb{E}[\hat{m}(x^*)] - m_{c^*}(z) = \underbrace{\left(\mathbb{E}[\hat{m}(x^*)] - m_W(z; x^*)\right)}_{\text{polynomial bias (in } Z)} + \underbrace{\left(m_W(z; x^*) - m_{c^*}(z)\right)}_{\text{Smoothing Bias (from } C)}.$$

The first term is the standard local-polynomial approximation error in Z , and satisfies

$$\mathbb{E}[\hat{m}(x^*)] - m_W(z; x^*) = \mathcal{O}(\|H\|^{p+1}).$$

282 Hence,

$$\mathbb{E}[\hat{m}(x^*)] - m_{c^*}(z) = O(\|H\|^{p+1}) + (m_W(z; x^*) - m_{c^*}(z)).$$

283 In particular, if the context kernels enforce exact selection of the context cell containing c^* (e.g.,
 284 $W(x^*, X) = 0$ outside that cell, so that $m_W(z; x^*) = m_{c^*}(z)$), then the smoothing bias term van-
 285 ishes and the $O(\|H\|^{p+1})$ rate is preserved. In the continuous-context case, sharpening the context
 286 kernels (smaller η) can reduce $|m_W(z; x^*) - m_{c^*}(z)|$, typically at the cost of increased variance
 287 through a smaller effective sample size. The displayed $O(\|H\|^{p+1})$ rate is a generic nonsharp
 288 umbrella upper bound inherited from the effective local polynomial problem; sharper interior
 289 rates depend on the usual parity refinements and are not pursued here.

290 Proof in [Appendix A.3](#).

291 **Proposition 2** (Asymptotic variance). Under Assumptions [1–5](#), fix $x^* \in \mathcal{X}$ such that $z = \psi_0(x^*)$
 292 is an interior point of the support of Z . Then

$$\text{Var}\{\hat{m}(x^*)\} = \frac{1}{n|H|} \frac{1}{f_Z(z)} e_1^\top M_0^{-1} \Omega_0 M_0^{-1} e_1 \times \Phi(x^*) + o((n|H|)^{-1}),$$

293 where M_0 and Ω_0 are the kernel moment matrices from Assumption [2](#), and

$$\Phi(x^*) = \frac{\mathbb{E}[W(x^*, X)^2 (Y - m_W(Z; x^*))^2 \mid Z = z]}{\mathbb{E}[W(x^*, X) \mid Z = z]^2}$$

294 is the effective residual factor. Under the model $Y = m_C(Z) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid Z, C] = 0$,

$$\Phi(x^*) = \frac{\mathbb{E}[W(x^*, X)^2 \{\sigma^2(z, C) + (m_C(z) - m_W(z; x^*))^2\} \mid Z = z]}{\mathbb{E}[W(x^*, X) \mid Z = z]^2}.$$

295 Proof in [Appendix A.4](#). The effective residual factor $\Phi(x^*)$ quantifies the statistical cost of
 296 context isolation and context mixing:

- 297 • **Recovery of standard LPR:** If $W \equiv 1$ (no context weights) and errors are homoskedastic
 298 and the model reduces to $Y = m(Z) + \varepsilon$, then $\Phi(x^*) = \sigma^2$, recovering the standard LPR
 299 variance expression.
- 300 • **Context-mixing variation:** Even when $\sigma^2(z, C) = 0$, $\Phi(x^*)$ can remain positive if mul-
 301 tiple contexts contribute nontrivially to the target $m_W(\cdot; x^*)$ at a given $Z = z$, through the
 302 term $(m_C(z) - m_W(z; x^*))^2$.
- 303 • **Weight-induced inflation:** When context weights are sparse or highly variable (e.g., fo-
 304 cusing on a small manifold slice), the squared-weight numerator relative to $\mathbb{E}[W \mid Z = z]^2$
 305 can inflate variance, especially when the effective residual variation is roughly homoge-
 306 neous or only weakly related to the weights. This reflects the reduction in effective sample
 307 size caused by the context filter. GC-LPR therefore does not eliminate the curse of di-
 308 mensionality; rather, it trades model bias against variance. While the method allows for
 309 complex dependencies, a low-variance estimate still requires a sufficiently dense sample
 310 in the joint (Z, C) space.

311 **Corollary 1** (Consistency for the context-smoothed target). *Under Assumptions 1–5, fix x^* such*
 312 *that $z = \psi_0(x^*)$ is an interior point of the support of Z . Then*

$$\mathbb{E}[(\hat{m}(x^*) - m_W(z; x^*))^2] = O(\|H\|^{2(p+1)}) + O((n|H|)^{-1}).$$

313 *Consequently, if $\|H\| \rightarrow 0$ and $n|H| \rightarrow \infty$, then*

$$\hat{m}(x^*) \xrightarrow{L^2} m_W(z; x^*), \quad \hat{m}(x^*) \xrightarrow{P} m_W(z; x^*).$$

314 *Proof.* Proposition 1 gives $\mathbb{E}[\hat{m}(x^*)] - m_W(z; x^*) = O(\|H\|^{p+1})$, while Proposition 2 and Assump-
 315 tions 4–5 give $\text{Var}\{\hat{m}(x^*)\} = O((n|H|)^{-1})$; combine these with $\mathbb{E}[(\hat{m} - m_W)^2] = \text{Var}(\hat{m}) + (\mathbb{E}[\hat{m}] -$
 316 $m_W)^2$. $\square$

317 Taken together, the results above give a pointwise-in- x^* i.i.d. asymptotic theory for the
 318 bandwidth-matrix formulation of GC-LPR around the induced target $m_W(z; x^*)$. Under $\|H\| \rightarrow 0$
 319 and $n|H| \rightarrow \infty$, Corollary 1 yields consistency for $m_W(z; x^*)$, but not, by itself, for the context-
 320 specific mean $m_{c^*}(z)$; the latter would require shrinking context bandwidths and regularity across
 321 contexts. The theory also does not cover the adaptive k -nearest-neighbor implementation used in
 322 the experiments below, nor the additional response-based robustness factor used in GRC-LPR.
 323 Extending the analysis to those settings is left for future work.

324 6. Applications and Experiments

325 The full experiment code, figure-generation scripts, and manuscript source are available in
 326 the GC-LPR repository at <https://github.com/yaniv-shulman/gclpr>. All empirical re-
 327 sults use the public Python implementation in the `rsklpr` package (Shulman, 2026), using the
 328 version 2.0 series. The empirical goal here is to compare GC/GRC-LPR against feature-only lo-
 329 cal polynomial smoothers and simple generic baselines, rather than to claim a benchmark against
 330 all task-specific spatial or graph regression model classes.

331 6.1. Experiment 1: Geospatial Context on California Housing

332 As a foundational validation of the proposed framework, we test the hypothesis that explicit
 333 context kernels improve performance even when contextual variables (coordinates) are already
 334 present in the primary feature set. This is intended as a within-family comparison against feature-
 335 only local smoothers rather than a full ablation of every geographic representation choice. We
 336 evaluate GC-LPR against standard LPR, its robust variant RSKLPR (Shulman, 2025), and a K-
 337 Nearest Neighbors (KNN) baseline.

338 **Dataset and Setup.** We utilize the California Housing dataset (Pace and Barry, 1997), modeling
 339 the median house value (`MedHouseVal`) using a feature vector Z containing `MedInc`, `AveRooms`,
 340 `Latitude`, and `Longitude`. Non-geospatial features were standardized. We use a repeated
 341 hold-out protocol with 5 repetitions of an 80%/20% train/test split and report mean and standard
 342 deviation over the held-out test results.

343 For each repetition, hyperparameters are selected by 4-fold cross-validation on the training
 344 split only, and the selected model is then evaluated once on the held-out test split. The hyper-
 345 parameter search space for local polynomial models included neighborhood size ($k \in [7, 151]$),
 346 polynomial degree ($p \in \{0, 1\}$), Laplacian and Tricube predictor kernels, and Minkowski or Ma-
 347 halanobis distances. For GC-LPR and GRC-LPR, the search also optimized the length-scale of

the Haversine context kernel. Additional evaluation details are collected in [Appendix B](#). The primary comparison is between:

- **LPR / RSKLPR (Standard):** Utilizes a single features kernel over the full feature vector Z .
- **GC-LPR / GRC-LPR (Geospatial):** Employs the compound kernel $K_c = K_{\text{std}}(Z) \cdot K_{\text{geo}}(\text{Lat}, \text{Lon})$, where K_{geo} is a Haversine-based RBF kernel.

Results and Statistical Analysis. The results are summarized in [Table 1](#). Both context-aware models outperform the feature-only local smoothers and KNN. GRC-LPR (the robust, context-aware variant) achieves the best mean performance across all three reported metrics, with RMSE 0.4993 ± 0.0074 , MAE 0.3324 ± 0.0034 , and $R^2 = 0.8134 \pm 0.0054$. GC-LPR is the second-best model, while the feature-only LPR family remains clearly ahead of KNN. The predictions scatter plot is shown in [Figure 1](#), and the split-wise RMSE distributions are shown in [Figure 2](#).

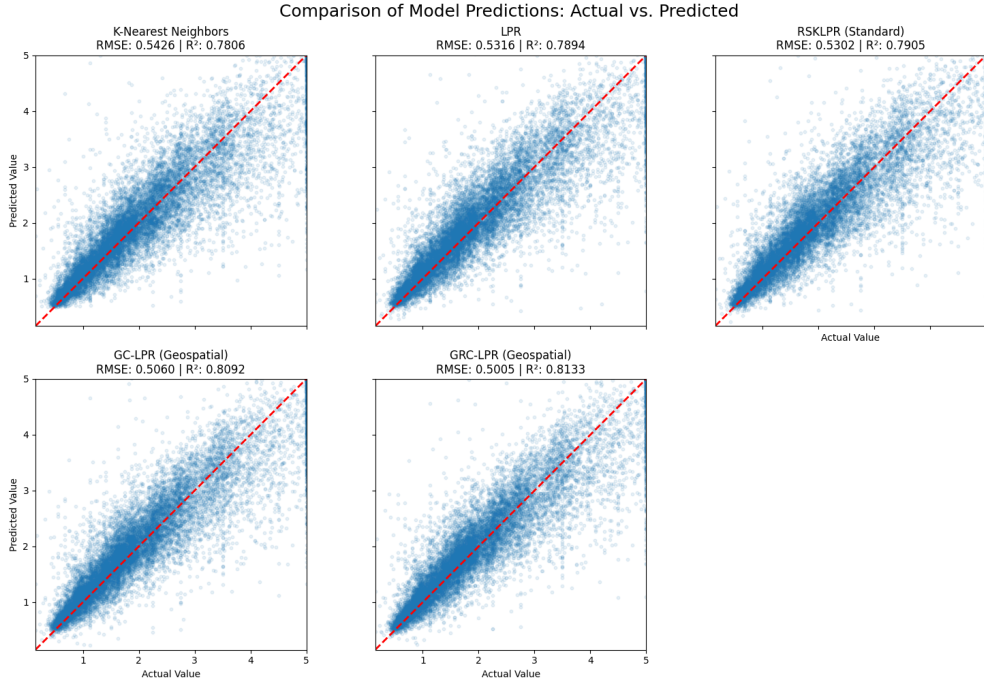


Figure 1: Comparison of model predictions for Experiment 1, aggregating the held-out predictions across the 5 repeated 80/20 train/test splits. The y-axis is the predicted value and x-axis is the actual value. The dashed line represents ideal predictions. The context-aware variants show tighter clustering around the ideal line than the feature-only smoothers.

The explicit decoupling of geospatial context leads to a marked improvement over standard LPR (RMSE 0.5060 vs. 0.5280). Incorporating the response-based robustness factor (K_r) improves the context-aware model further, reducing RMSE from 0.5060 to 0.4993 and MAE from 0.3398 to 0.3324. In contrast, the robustness factor alone offers only a modest gain in the feature-only setting (RSKLPR vs. LPR).

Table 1: Experiment 1 results on the California Housing dataset under 5 repeated 80/20 hold-out splits with 4-fold cross-validation on each training split. Values are mean $\pm$ standard deviation over held-out test metrics.

Model	RMSE	MAE	R^2
GRC-LPR (Geospatial)	0.4993 ± 0.0074	0.3324 ± 0.0034	0.8134 ± 0.0054
GC-LPR (Geospatial)	0.5060 ± 0.0098	0.3398 ± 0.0051	0.8084 ± 0.0067
RSKLPR (Standard)	0.5267 ± 0.0100	0.3501 ± 0.0055	0.7923 ± 0.0090
LPR (Standard)	0.5280 ± 0.0066	0.3536 ± 0.0040	0.7914 ± 0.0063
KNN	0.5423 ± 0.0117	0.3562 ± 0.0055	0.7798 ± 0.0118

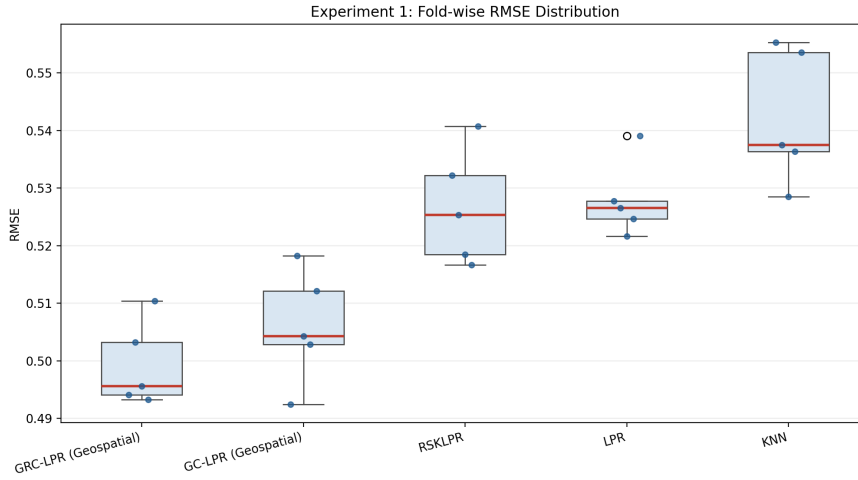


Figure 2: RMSE distributions for Experiment 1 under the repeated 80/20 hold-out protocol. The context-aware variants shift the full RMSE distribution downward, with GRC-LPR achieving the lowest center and the strongest average performance across held-out splits.

365 **Spatial Error Distribution.** To investigate the qualitative impact of context-awareness, we mapped
366 the prediction errors across California (Figure 3). The bottom row, corresponding to GC-LPR
367 and GRC-LPR, is visibly more uniform than the feature-only local smoothers in the top row.
368 The context-aware models better adapt to the distinct price dynamics of the San Francisco Bay
369 Area and Greater Los Angeles, whereas the feature-only smoothers still over-smooth across these
370 spatial boundaries.

371 **Effective Neighborhood Analysis.** Across the 5 repeated hold-out runs, the selected neighbor-
372 hoods remained noticeably larger for the context-aware variants than for the feature-only local
373 smoothers: $k \in \{117, 127, 137\}$ for GC-LPR and $k \in \{117, 137, 147\}$ for GRC-LPR, compared
374 with $k \in \{57, 67\}$ for LPR and $k \in \{27, 37\}$ for RSKLPR. The local-polynomial models uniformly
375 selected $p = 1$. This is consistent with the intended role of the context kernel: it allows the
376 estimator to borrow strength from a wider Euclidean neighborhood while still down-weighting
377 geographically irrelevant observations. The result is a more stable local fit without the same
378 degree of cross-region smoothing bias.

Geospatial Analysis of Model Performance (Test Set Errors)

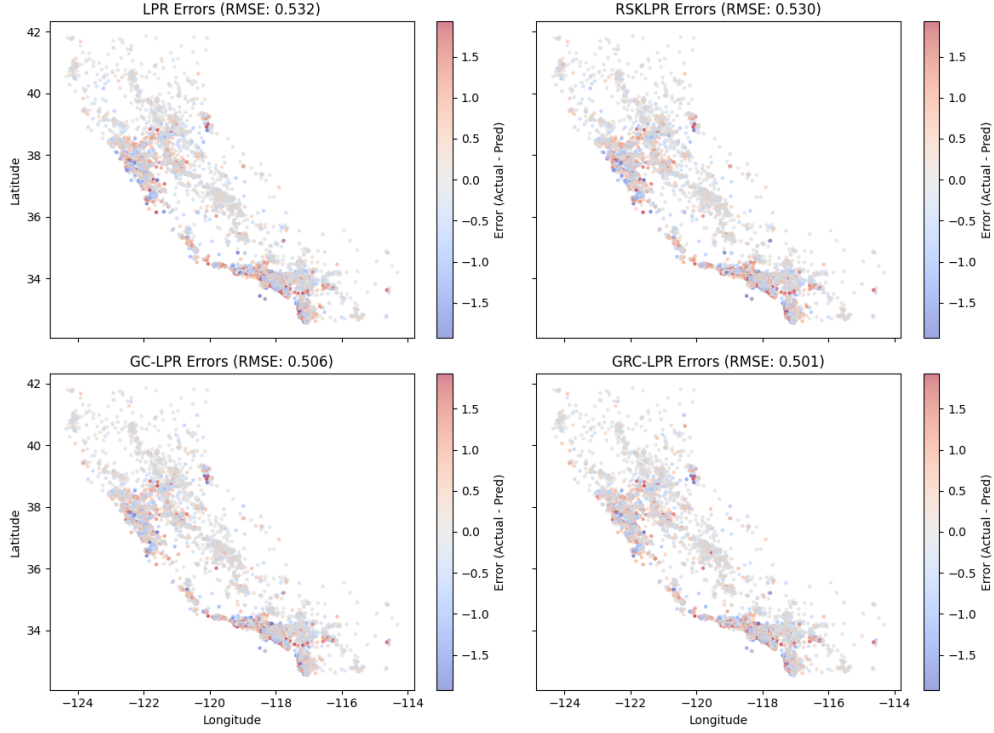


Figure 3: Geospatial distribution of prediction errors across California, aggregating the held-out predictions across the 5 repeated 80/20 train/test splits. The top row shows feature-only local smoothers (LPR and RSKLPR), while the bottom row shows the context-aware variants (GC-LPR and GRC-LPR). The context-aware models exhibit lower and more spatially uniform errors, especially in high-variance coastal urban centers.

6.2. Experiment 2: Fusing Tabular, Geospatial, and Network Context in NYC

The purpose of the second experiment is to demonstrate the use of a more advanced geospatial kernel that takes similar inputs as in Experiment 1 (i.e., Latitude and Longitude), but employs a non-Euclidean kernel on a graph. The goal is to show that standard distance metrics (Euclidean or Haversine) are often insufficient, and that a context-aware model can capture a more meaningful, network-based definition of proximity.

Dataset and Setup. We use the archived NYC detailed-listings snapshot dated February 13, 2026 from Inside Airbnb ([Inside Airbnb, 2026](#)) to predict listing price. The features include standard tabular attributes (`host_listings_count`, `room_type`, etc.) as well as Latitude and Longitude. The subway context graph is reconstructed from the MTA New York City Transit static GTFS subway feed ([Metropolitan Transportation Authority, 2026](#)). We use a repeated hold-out protocol: for each of 5 repetitions, the data are split into 80% training and 20% test sets, hyperparameters are selected by 4-fold cross-validation on the training split, and the selected model is then evaluated once on the held-out test split. The local models are fit on

393 $\log(1 + \text{price})$, but tuning and reported metrics are based on raw-price RMSE after inverse
 394 transformation. For the local smoothers, the neighborhood search is restricted to the stable local
 395 range $k \in \{12, \dots, 20\}$. Table 2 reports the mean and standard deviation across the 5 held-out test
 396 results. Additional implementation details are collected in Appendix B. The primary comparison
 397 is between:

- 398 • **KNN**: A distance-weighted k -nearest-neighbor baseline on the same tabular and geospatial
 399 predictors.
- 400 • **LPR / RSKLPR (Standard)**: Feature-only local polynomial smoothers using the standard
 401 predictors, with RSKLPR adding the response-based robustness factor.
- **GC-LPR / GRC-LPR (Subway)**: Context-aware local smoothers that combine a com-
 pound predictor kernel over the standard predictors and the subway network context, with
 GRC-LPR additionally including the response-based robustness factor:

$$K_c = K_{\text{features}} \times K_{\text{subway}}, \quad \mathcal{K}_{\mathcal{D}}^* = K_c \times K_r$$

- 402 – K_{features} : A kernel on the standard tabular features.
- 403 – K_{subway} : A dedicated context kernel. This kernel maps coordinates to the nearest
 404 subway station and computes similarity based on the shortest path distance (number
 405 of stops) between stations on the subway network graph. This captures topological
 406 proximity rather than simple aerial distance. Figure 5 illustrates the spatial decay
 407 of similarity weights on the subway graph, demonstrating how the kernel captures
 408 non-Euclidean connectivity relative to a central station.
- 409 – K_r : The response-based robustness factor from RSKLPR.

410 **Results.** The repeated hold-out results in Table 2 show that the subway-context kernel improves
 411 on the feature-only LPR family and also edges out KNN on the primary RMSE metric. GRC-
 412 LPR (Subway) achieves the best mean RMSE and mean R^2 , with GC-LPR (Subway) a close sec-
 413 ond, while KNN attains the best mean MAE. The feature-only local smoothers trail the context-
 414 aware variants by a substantial margin. Figure 4 shows the RMSE distributions across the 5
 415 held-out splits and confirms that the subway-context models remain competitive across repeats
 416 rather than only on a single favorable split.

417 **Analysis of Hyperparameters.** The selected hyperparameters again favor very local smoothing.
 418 Across the 5 repeated hold-out runs, the subway-context models consistently selected $p = 0$,
 419 neighborhood sizes between $k = 12$ and $k = 14$, and graph distance scales of 1.0 or 1.5. LPR
 420 likewise remained highly local, usually selecting $k = 12$ or $k = 13$, while KNN selected between
 421 $k = 3$ and $k = 17$ depending on the split. Unlike Experiment 1, the context kernel does not en-
 422 courage substantially larger neighborhoods. Instead, the selected subway-context models remain
 423 in the small- k regime, which is consistent with a highly irregular price surface where the graph
 424 context sharpens locality rather than broadening it.

425 6.3. Experiment 3: Regression on a Node Function Defined over a Graph

426 This experiment serves as the primary validation of our theoretical framework. While the
 427 previous experiments relied on real-world data where the true data generating process is un-
 428 known, here we explicitly simulate a target function $Y = m_C(Z)$ defined on the nodes of a graph.

Table 2: Model performance on the NYC Airbnb data under the repeated 80/20 hold-out protocol. Entries report mean $\pm$ standard deviation across 5 held-out test splits.

Model	RMSE (Price)	MAE (Price)	R^2
GRC-LPR (Subway)	\$1660.35 $\pm$ 407.59	\$213.53 $\pm$ 35.05	0.8631 $\pm$ 0.0554
GC-LPR (Subway)	\$1693.22 $\pm$ 417.79	\$218.58 $\pm$ 34.94	0.8574 $\pm$ 0.0583
KNN	\$1727.02 $\pm$ 332.86	\$208.93 $\pm$ 28.36	0.8537 $\pm$ 0.0482
RSKLPR (Standard)	\$1995.98 $\pm$ 301.63	\$240.79 $\pm$ 30.67	0.8056 $\pm$ 0.0483
LPR (Standard)	\$2029.73 $\pm$ 323.32	\$244.34 $\pm$ 32.62	0.7988 $\pm$ 0.0536

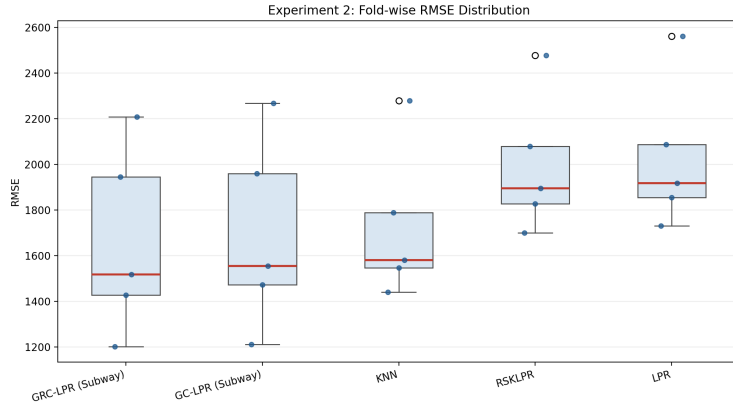


Figure 4: RMSE distributions for Experiment 2 under the repeated 80/20 hold-out protocol. The subway-context models improve on the feature-only LPR variants across repeated held-out test splits, with GRC-LPR (Subway) and GC-LPR (Subway) forming the leading pair.

This allows us to test the specific hypothesis that *structural locality* (who you are connected to) is distinct from *feature similarity* (how connected you are), and that omitting the former leads to bias.

Dataset and Setup. We construct a graph $G = (\mathcal{V}, \mathcal{E})$ representing the US domestic flight network from the public `airports.csv` airport-metadata file and `flights-airport.csv` route-count file distributed with Vega’s airport-connections example (Vega, 2026a,b), where the route counts correspond to year 2008.

- **Node Features (Z):** For each airport, we compute a feature vector containing standard graph centrality metrics (PageRank, Betweenness, Degree) and geospatial coordinates (Latitude, Longitude). Crucially, these features describe the *role* of a node, but not its unique position in the topology.
- **Generative Process (Y):** We generate a synthetic “flight delay” signal that depends on both local features and network propagation. Let $\widehat{\text{PR}}_i$, $\widehat{\text{BW}}_i$, and $\widehat{\text{Deg}}_i$ denote the standardized PageRank, betweenness, and degree features of node i , and let $\xi_i \stackrel{\text{iid}}{\sim} N(0, 1)$. The base

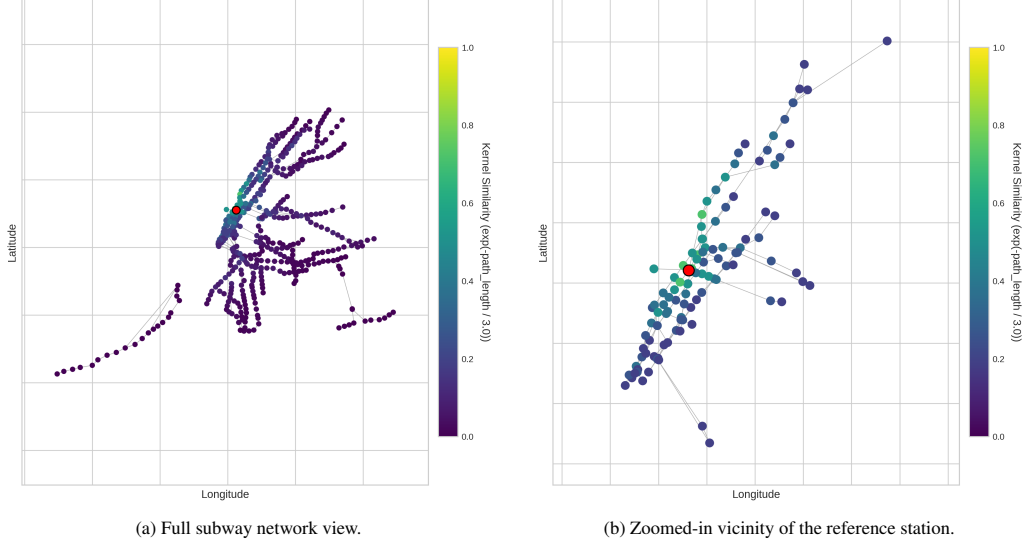


Figure 5: Visualizing graph-based kernel similarity for Experiment 2. The heatmaps demonstrate the similarity weights generated by K_{subway} relative to the reference station “Times Sq-42 St” (ID 127), indicated by the large red node with a black outline. Warmer colors represent higher similarity, which decays exponentially based on the shortest path distance on the subway graph network. Panel (a) shows the distribution over the entire network, while (b) highlights the local neighborhood topology.

signal is

$$\begin{aligned}\tilde{Y}_i^{(0)} &= 0.3 \widetilde{\text{PR}}_i + 0.5 \widetilde{\text{BW}}_i + 0.2 \widetilde{\text{Deg}}_i + 0.1 \xi_i, \\ \tilde{Y}^{(k+1)} &= 0.7 \tilde{Y}^{(0)} + 0.3 A \tilde{Y}^{(k)} \quad \text{for } k = 0, \dots, 6,\end{aligned}$$

where A is the row-normalized traffic-weighted adjacency matrix and the final target is $Y := \tilde{Y}^{(7)}$, shifted to remain strictly positive. For clarity, the synthetic propagation uses this traffic-weighted normalized adjacency, whereas the GC/GRC-LPR context kernel itself is built on the underlying undirected airport graph via a symmetric shortest-path similarity. Further implementation details are collected in [Appendix B](#).

This process creates a scenario where two airports with similar feature vectors Z (e.g., two medium-sized hubs) may have vastly different Y values if they lie in different diffusion clusters. Standard LPR with a kernel only in Z cannot distinguish them; GC-LPR with graph context can. Figure 6 illustrates the underlying network topology and the synthetic delay function used as the prediction target.

Models. We compare the same local-regression family used in the earlier experiments: KNN, LPR, RSKLPR, GC-LPR (Graph), and GRC-LPR (Graph). Experiment 3 uses 5 repeated 80/20 hold-out splits. For each repetition, hyperparameters are selected by 4-fold cross-validation on the training split, and Table 3 reports the mean and standard deviation of the held-out test metrics across the 5 repetitions.

- **Feature-only baselines:** LPR and RSKLPR use a tricube kernel over Z with the Mahalanobis metric. Across the 5 repeated hold-out runs, the selected neighborhood sizes fall

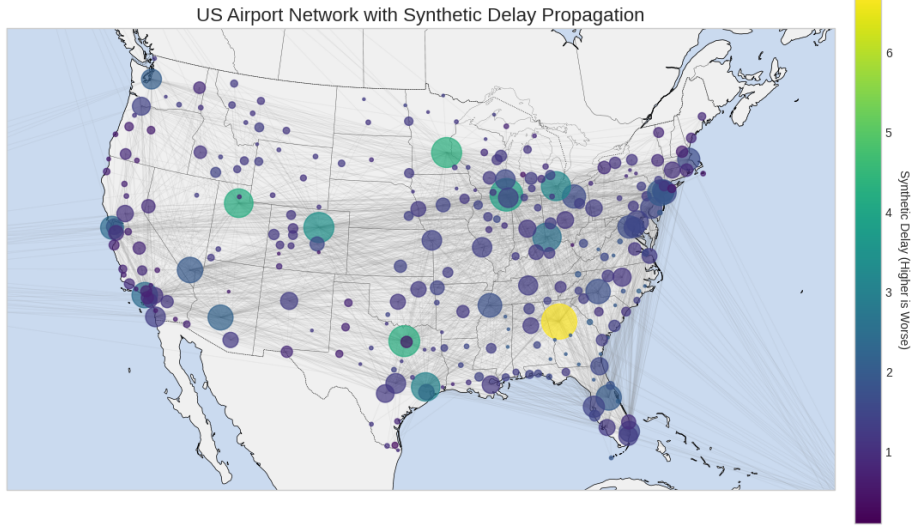


Figure 6: Visualizing the experimental setup for Experiment 3. The map displays the US airport network where edges represent flight routes and node sizes are proportional to degree. The color gradient shows the synthetic “delay” target function (Y) propagated across the network, which serves as the ground truth for the regression task.

in the range **79–111** with polynomial degree fixed at $p = 1$. KNN remains as a generic local nonparametric baseline.

- **Graph-context models:** Our proposed graph-aware variants utilize the compound predictor kernel

$$K_c(x, X_i) = K_{\text{feat}}(Z_i - Z_x) \times K_{\text{graph}}(C_i, C_x)$$

and, when the response-based robustness factor is used,

$$\mathcal{K}_{\mathcal{D}}^* = K_c \times K_r.$$

Here, K_{graph} is based on *unweighted* shortest-path distance on the airport graph. Across the repeated hold-out runs, GC/GRC-LPR select neighborhood sizes between **111** and **135**, with graph distance scales concentrated between **0.35** and **0.40** (and one run at 0.75). As shown in Figure 7, the graph kernel assigns high weights to airports that are close in hop distance on the route network, thereby defining a non-Euclidean local neighborhood.

Results. The repeated held-out results in Table 3 show that the graph-context models consistently improve on feature-only local smoothing. Because the target is generated by graph propagation on the same underlying network later used to define the context kernel, this experiment should be read as a mechanism check for graph-structured dependence rather than as a broad benchmark against all graph regression methods.

The performance margins are modest but consistent. GC-LPR (Graph) attains the best mean RMSE and mean R^2 , while GRC-LPR (Graph) is effectively tied and slightly improves mean MAE only negligibly. Relative to feature-only LPR, the graph kernel lowers mean RMSE from 0.2127 to 0.2085 and improves mean R^2 from 0.8853 to 0.8901. The weak gain from the

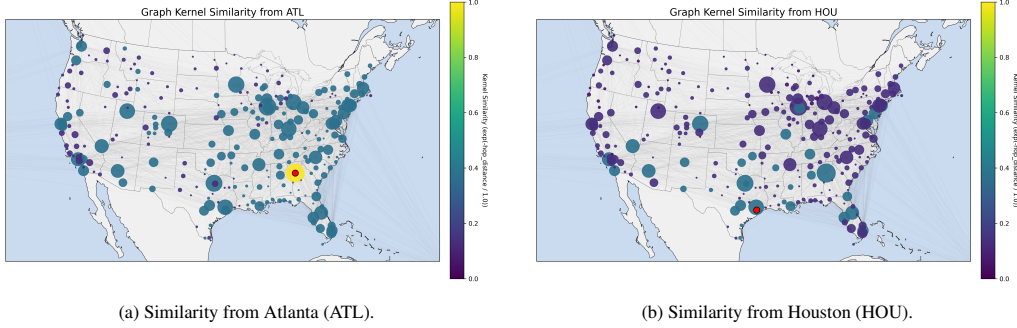


Figure 7: Visualization of the graph kernel K_{graph} used in the final Experiment 3 specification. The heatmaps show similarity based on unweighted shortest-path (hop) distance on the route network rather than Euclidean distance. The red nodes indicate the query reference airports (ATL and HOU).

Table 3: Experiment 3 performance under 5 repeated 80/20 hold-out splits with 4-fold cross-validation on each training split. Values are mean $\pm$ standard deviation over held-out test metrics.

Model	RMSE	MAE	R^2
GC-LPR (Graph)	0.2085 $\pm$ 0.0264	0.1529 $\pm$ 0.0169	0.8901 $\pm$ 0.0420
GRC-LPR (Graph)	0.2087 $\pm$ 0.0267	0.1532 $\pm$ 0.0166	0.8897 $\pm$ 0.0430
RSKLPR	0.2124 $\pm$ 0.0217	0.1590 $\pm$ 0.0158	0.8857 $\pm$ 0.0454
LPR	0.2127 $\pm$ 0.0218	0.1595 $\pm$ 0.0156	0.8853 $\pm$ 0.0458
KNN	0.4326 $\pm$ 0.1072	0.2586 $\pm$ 0.0218	0.5573 $\pm$ 0.1079

response-based robustness factor indicates that, in this synthetic setting, the main improvement comes from incorporating graph topology into the predictor kernel rather than from additional response reweighting.

6.4. Experiment 4: Public Graph-Signal Forecasting on the Hungary Chickenpox Benchmark

To complement the synthetic airport study with an external public benchmark, we next evaluate GC-LPR on the Chickenpox Cases in Hungary benchmark of [Rozemberczki et al. \(2021\)](#). This is a fixed graph-signal regression problem: the graph consists of 20 counties connected by 61 undirected edges, and the response is the weekly chickenpox count observed at each county over 521 weeks. After constructing four lagged predictors from the past weekly counts, each county-time pair becomes one regression sample, yielding 10,340 node-time observations with

$$Z = (\text{lag_4}, \text{lag_3}, \text{lag_2}, \text{lag_1})$$

and target Y equal to the next week’s count.

Dataset and Setup. This benchmark differs from Experiments 1–3 in that the graph and response are both observed rather than simulated, and the train/test protocol must preserve temporal ordering. Accordingly, Experiment 4 uses a rolling-origin design with 5 outer chronological train/test splits. On each outer split, hyperparameters are selected by 4-fold time-series cross-validation on the training window only, and the selected model is then evaluated on the next contiguous test block. Figures 8–10 summarize the held-out predictions, county-level error structure, and fold-wise RMSE distributions.

498 **Models.** We again compare KNN, LPR, RSKLPR, GC-LPR (Graph), and GRC-LPR (Graph).
 499 The feature-only models operate only on the lagged-count vector Z using a tricube feature kernel
 500 and Manhattan distance. The graph-aware models use the compound kernel

$$K_c(x, X_i) = K_{\text{feat}}(Z_i - Z_x) \times K_{\text{graph}}(C_i, C_x),$$

501 where K_{graph} is an exponentially decaying hop-distance kernel on the county graph. The selected
 502 hyperparameters are highly stable across the 5 outer splits: LPR and RSKLPR always choose
 503 **size_neighborhood** = 121, while GC/GRC-LPR always choose **size_neighborhood** = 361 and
 504 **distance_scale** = 16.0. This indicates that, on this benchmark, graph context is useful but acts as
 505 a broad structural regularizer rather than as a sharply localized graph slice.

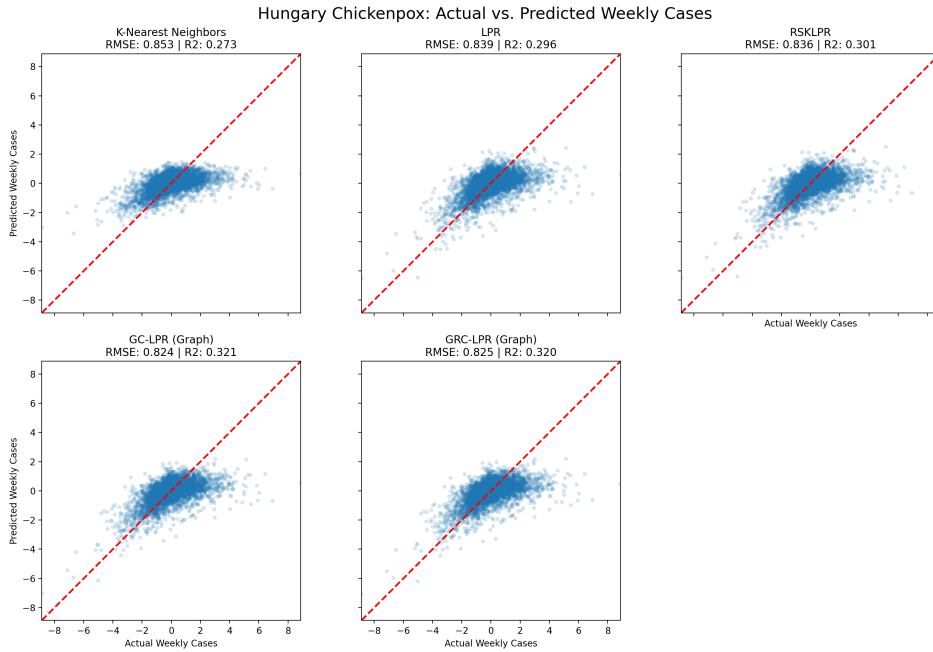


Figure 8: Experiment 4 held-out prediction scatter plots aggregated over the 5 rolling-origin test windows. Each panel compares predicted and observed county-week chickenpox counts for one model.

506 **Results.** Table 4 reports the mean and standard deviation of the held-out test metrics across the
 507 5 rolling-origin splits. The gains from graph context are modest but consistent. GC-LPR (Graph)
 508 attains the best mean RMSE and mean R^2 , while GRC-LPR (Graph) attains the best mean MAE
 509 by a very small margin. Relative to feature-only LPR, the hop-based graph kernel lowers mean
 510 RMSE from 0.8347 to 0.8201 and improves mean R^2 from 0.2961 to 0.3205.

511 Taken together, Experiments 3 and 4 show two complementary graph-context regimes. In
 512 the synthetic airport task, the graph kernel recovers an explicitly simulated topological signal.
 513 On the public chickenpox benchmark, the benefit is smaller but still systematic, which is the
 514 more relevant result for external validation. In both cases, the main improvement comes from
 515 incorporating graph structure into the predictor kernel; the additional response-based robustness
 516 factor provides only marginal gains.

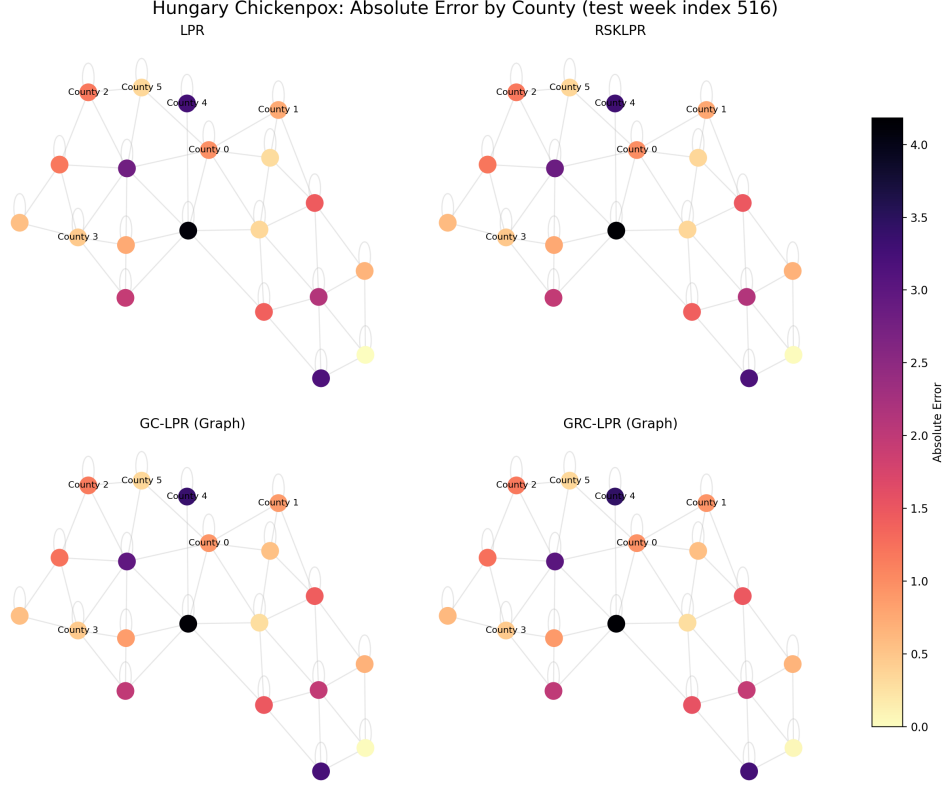


Figure 9: Experiment 4 county-level absolute-error maps on the Hungary chickenpox graph. Colors summarize held-out prediction error aggregated over the rolling-origin evaluation windows.

7. Conclusion

This work presented Generalized Context-Aware Local Polynomial Regression (GC-LPR), a framework that resolves a fundamental rigidity in classical nonparametric smoothing: the requirement that the variables used to define the local neighborhood must be identical to those used in the polynomial fit. By decomposing the predictor space into a Euclidean fitting chart ψ_0 with coordinates $Z = \psi_0(X)$ and a context space (C), and bridging them via a compound product kernel, we have established a method that is both mathematically rigorous and practically flexible.

Our theoretical analysis re-frames the role of "context" not merely as a variance modifier, but as a fundamental component of the generative process $Y = m_C(Z)$. In the fixed-context-bandwidth asymptotics developed here, GC-LPR acts as a projected smoother: the context kernels (whether discrete, geospatial, or graph-based) isolate a specific slice of the data manifold, effectively "masking" irrelevant observations before the local polynomial estimation takes place in Z . The resulting bias-variance theory is therefore for the induced context-smoothed target m_W , not automatically for the point target $m_{c^*}(z)$. This interpretation justifies the use of kernels that are discontinuous with respect to Euclidean distance, such as those defined on road networks or administrative partitions.

Table 4: Experiment 4 performance on the public Hungary chickenpox benchmark under 5 rolling-origin splits with 4-fold time-series cross-validation on each training window. Values are mean $\pm$ standard deviation over held-out test metrics.

Model	RMSE	MAE	R^2
GC-LPR (Graph)	0.8201 ± 0.0906	0.5370 ± 0.0569	0.3205 ± 0.0407
GRC-LPR (Graph)	0.8212 ± 0.0904	0.5367 ± 0.0572	0.3187 ± 0.0408
RSKLPR	0.8320 ± 0.0933	0.5458 ± 0.0604	0.3007 ± 0.0441
LPR	0.8347 ± 0.0952	0.5490 ± 0.0613	0.2961 ± 0.0473
KNN	0.8486 ± 0.0976	0.5557 ± 0.0608	0.2736 ± 0.0321

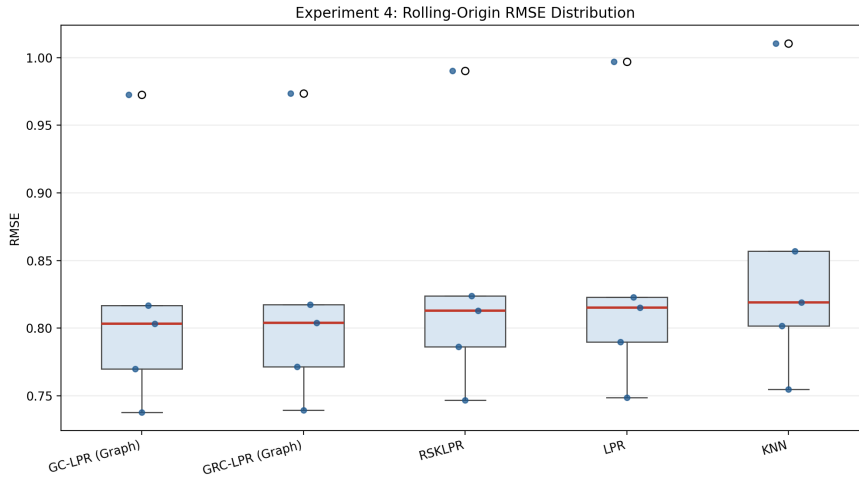


Figure 10: Fold-wise RMSE distribution for Experiment 4 under the rolling-origin protocol. The graph-aware models shift the RMSE distribution downward relative to KNN and feature-only LPR variants.

Empirically, we demonstrated that this decoupling yields significant performance gains. In the California Housing experiment, we showed that explicit geospatial kernels improve accuracy even when coordinates are already included as features. In the US Airport Network experiment, we demonstrated that topological contexts (flight paths) capture dependencies that standard Euclidean metrics strictly miss. On the public Hungary chickenpox benchmark, graph-aware local smoothers again improved on feature-only baselines, albeit with smaller margins, providing external validation beyond the synthetic graph experiment.

Ultimately, GC-LPR bridges the gap between interpretable classical statistics and modern geometric learning. It offers a powerful tool for domains where data is tabular but lives on an irregular domain, providing the adaptivity of graph signal processing with the well-understood local-polynomial bias-variance analysis around the induced target m_W .

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637 Appendix A. Proofs

638 In this section, we prove the main population-identification and asymptotic results. The
 639 derivations follow the standard local polynomial regression arguments (Fan and Gijbels, 1996)
 640 while explicitly tracking the influence of the compound context weight $W(x, u)$.

641 Appendix A.1. Notation and Setup

642 For simplicity of notation, throughout this appendix we define the chart coordinates used for
 643 the polynomial fit as $Z_i := \psi_0(X_i)$ and the query point as $z := \psi_0(x)$. The polynomial basis and
 644 the kernel K_0 operate on Z , while the context weights W depend on the full predictors X .

645 The main text defines the estimator using the unscaled basis $r_p(Z_i - z)$. For asymptotic cal-
 646 culations, it is convenient to use the equivalent scaled basis $r_p(t_i)$, where

$$t_i := H^{-1}(Z_i - z).$$

647 Because composing a polynomial of total degree at most p with the linear map H stays in the
 648 same polynomial space, there exists an invertible matrix A_H such that

$$r_p(Z_i - z) = A_H r_p(t_i).$$

649 Hence the main-text objective can be reparameterized by a coefficient vector $\vartheta(x) = A_H^\top \beta(x)$, and
 650 the intercept is unchanged because the constant basis term is invariant:

$$\hat{m}(x) = e_1^\top \hat{\beta}(x) = e_1^\top \hat{\vartheta}(x).$$

651 The scaled-basis estimator at a query point x is:

$$\hat{\vartheta}(x) = \arg \min_{\vartheta} \sum_{i=1}^n \left(Y_i - r_p(t_i)^\top \vartheta \right)^2 K_0(t_i) W(x, X_i),$$

652 where $W(x, X_i) = \prod_{j=1}^J K_j(\psi_j(x), \psi_j(X_i))$ collects all context weights.

653
 654 Let R be the $n \times M$ design matrix with rows $r_p(t_i)^\top$, and let the weight matrix be $W_x =$
 655 $\text{diag}\{K_0(t_i) W(x, X_i)\}$. The weighted least squares solution is:

$$\hat{\vartheta}(x) = (R^\top W_x R)^{-1} R^\top W_x Y.$$

656 We define the local moment matrix S_n and the projection vector T_n :

$$S_n = \frac{1}{n|H|} R^\top W_x R, \quad T_n = \frac{1}{n|H|} R^\top W_x Y.$$

657 Conditioning on the predictors $\mathbb{X} = \{X_1, \dots, X_n\}$, the expectation is:

$$\mathbb{E}[\hat{\vartheta}(x) \mid \mathbb{X}] = S_n^{-1} \mathbb{E}[T_n \mid \mathbb{X}] = S_n^{-1} \left(\frac{1}{n|H|} R^\top W_x \mathbf{m} \right),$$

658 where $\mathbf{m} = (m_{C_1}(Z_1), \dots, m_{C_n}(Z_n))^\top$.

659 *Appendix A.2. Proof of Lemma 1 (Context-smoothed conditional target)*

660 Because Lemma 1 is the population identification step underlying the later bias analysis, we
661 record a slightly expanded derivation.

662 *Proof.* Fix $x \in \mathcal{X}$ and $u \in \mathbb{R}^d$, and write

$$m_W := m_W(u; x), \quad \gamma := \gamma_x(u).$$

663 Throughout this proof, u is fixed, so α and m_W are ordinary scalars in the conditional problem
664 given $Z = u$. Define

$$L_u(\alpha) := \mathbb{E}[(Y - \alpha)^2 W(x, X) \mid Z = u].$$

665 Under the lemma assumptions, $0 < \gamma < \infty$ and $\mathbb{E}[Y^2 W(x, X) \mid Z = u] < \infty$, so $L_u(\alpha)$ is finite for
666 each $\alpha \in \mathbb{R}$.

667 Now decompose

$$Y - \alpha = (Y - m_W) + (m_W - \alpha).$$

668 Therefore,

$$(Y - \alpha)^2 = (Y - m_W)^2 + 2(Y - m_W)(m_W - \alpha) + (m_W - \alpha)^2.$$

669 Multiplying by $W(x, X)$ and taking conditional expectations given $Z = u$, we obtain

$$\begin{aligned} L_u(\alpha) &= \mathbb{E}[(Y - m_W)^2 W(x, X) \mid Z = u] \\ &\quad + 2(m_W - \alpha) \mathbb{E}[(Y - m_W) W(x, X) \mid Z = u] \\ &\quad + (m_W - \alpha)^2 \mathbb{E}[W(x, X) \mid Z = u]. \end{aligned}$$

670 The cross-term vanishes because

$$\mathbb{E}[(Y - m_W) W(x, X) \mid Z = u] = \mathbb{E}[Y W(x, X) \mid Z = u] - m_W \mathbb{E}[W(x, X) \mid Z = u] = 0,$$

671 by the definition $m_W = \mathbb{E}[Y W(x, X) \mid Z = u] / \gamma$ and $\gamma = \mathbb{E}[W(x, X) \mid Z = u]$. Hence

$$L_u(\alpha) = \mathbb{E}[(Y - m_W)^2 W(x, X) \mid Z = u] + \gamma(\alpha - m_W)^2.$$

672 Since $\gamma > 0$, the right-hand side is uniquely minimized at $\alpha = m_W$, proving the scalar argmin
673 statement.

674 For the function version, let $a : \mathbb{R}^d \rightarrow \mathbb{R}$ be measurable. Under the conditional law given
675 $Z = u$, one has $a(Z) = a(u)$ almost surely. Applying the scalar identity with $\alpha = a(u)$ yields

$$\mathbb{E}[(Y - a(Z))^2 W(x, X) \mid Z = u] = \mathbb{E}[(Y - m_W(u; x))^2 W(x, X) \mid Z = u] + \gamma_x(u) (a(u) - m_W(u; x))^2.$$

676 This proves Lemma 1. □

677 *Appendix A.3. Proof of Proposition 1 (Bias decomposition and rate)*

678 Throughout the proof we use the notation of Appendix A.1.

679 *Proof.* Fix a query point $x^\star$ and write $z := \psi_0(x^\star)$. By Lemma 1, the context-smoothed mean
 680 $m_W(u; x^\star)$ is, for each u , the unique conditional weighted least-squares target induced by the
 681 context weight $W(x^\star, X)$. By Lemma 2, the population GC-LPR objective at $x^\star$,

$$\mathcal{J}_{\text{gc-lpr}}(x^\star; \beta),$$

682 is, up to an additive constant independent of β , exactly the standard local polynomial objective
 683 for the target $m_W(\cdot; x^\star)$ under the effective design density

$$u \mapsto f_Z(u) \gamma_{x^\star}(u).$$

684 Under Assumptions 1–4, this induced problem satisfies the usual local regularity conditions
 685 near z : $m_W(\cdot; x^\star) \in C^{p+1}$ by Assumption 1, f_Z is continuous with $f_Z(z) > 0$, $\gamma_{x^\star}(z) = \phi_1(x^\star) > 0$
 686 by Assumption 4, and $\gamma_{x^\star}$ is locally Lipschitz near z by Assumption 3. Therefore the standard
 687 local polynomial bias expansion applies to the intercept estimator for this effective problem (see,
 688 e.g., Fan and Gijbels (1996); Loader (1999)), yielding

$$\mathbb{E}[\hat{m}(x^\star)] - m_W(z; x^\star) = O(\|H\|^{p+1}).$$

689 Now add and subtract $m_W(z; x^\star)$:

$$\mathbb{E}[\hat{m}(x^\star)] - m_{c^\star}(z) = \left(\mathbb{E}[\hat{m}(x^\star)] - m_W(z; x^\star) \right) + \left(m_W(z; x^\star) - m_{c^\star}(z) \right).$$

690 The first term is the polynomial approximation bias in the Z -coordinates and is $O(\|H\|^{p+1})$; the
 691 second is the context-smoothing gap. Hence

$$\mathbb{E}[\hat{m}(x^\star)] - m_{c^\star}(z) = O(\|H\|^{p+1}) + \left(m_W(z; x^\star) - m_{c^\star}(z) \right).$$

692 If the context kernels enforce exact selection of the context cell containing $c^\star$, then $m_W(z; x^\star) =$
 693 $m_{c^\star}(z)$ and the smoothing term vanishes. In the continuous-context case, shrinking the context
 694 bandwidths sharpens the selection and can reduce $|m_W(z; x^\star) - m_{c^\star}(z)|$, at the cost of increased
 695 variance through a smaller effective sample size. This proves Proposition 1. $\square$

696 Appendix A.4. Proof of Proposition 2 (Asymptotic variance)

697 *Proof.* Fix $x^\star \in \mathcal{X}$ and write $z = \psi_0(x^\star)$. By Lemma 2, the variance analysis is that of standard
 698 local polynomial regression for the target $m_W(\cdot; x^\star)$ under the effective design density

$$q_{x^\star}(u) := f_Z(u) \gamma_{x^\star}(u).$$

699 Define the effective residual and its weighted conditional second moment:

$$\xi_{x^\star} := Y - m_W(Z; x^\star), \quad \nu_{x^\star}(u) := \mathbb{E}\left[W(x^\star, X)^2 \xi_{x^\star}^2 \mid Z = u\right].$$

700 By Lemma 1,

$$\mathbb{E}\left[\xi_{x^\star} W(x^\star, X) \mid Z = u\right] = 0,$$

701 and Assumptions 1–5 imply that, in a neighborhood of z , the effective design density $q_{x^\star}$ is
 702 continuous and bounded away from zero, while $\nu_{x^\star}$ is locally bounded and continuous at z .

703 Write $\xi_{i,x^\star} := Y_i - m_W(Z_i; x^\star)$ and decompose the score term as

$$T_n^\xi(x^\star) := \frac{1}{n|H|} \sum_{i=1}^n K_0(t_i) W(x^\star, X_i) r_p(t_i) \xi_{i,x^\star}.$$

704 By the same change-of-variables and dominated-convergence arguments as in standard LPR (us-
705 ing the continuity of $q_{x^\star}$ and $v_{x^\star}$, together with the moment conditions in Assumption 2), the
706 local moment matrix from Appendix A.1 satisfies

$$S_n \xrightarrow{P} q_{x^\star}(z) M_0,$$

707 and independence of $\{(X_i, Y_i)\}_{i=1}^n$ gives

$$\text{Var}(T_n^\xi(x^\star)) = \frac{1}{n|H|} f_Z(z) v_{x^\star}(z) \Omega_0 + o((n|H|)^{-1}).$$

708 Since the centered stochastic part of $\hat{\vartheta}(x^\star)$ is $S_n^{-1} T_n^\xi(x^\star)$, Slutsky's theorem yields

$$\text{Var}(\hat{\vartheta}(x^\star)) = \frac{1}{n|H|} \frac{v_{x^\star}(z)}{f_Z(z) \gamma_{x^\star}(z)^2} M_0^{-1} \Omega_0 M_0^{-1} + o((n|H|)^{-1}).$$

709 Taking the intercept component gives

$$\text{Var}(\hat{m}(x^\star)) = \frac{1}{n|H|} \frac{v_{x^\star}(z)}{f_Z(z) \gamma_{x^\star}(z)^2} e_1^\top M_0^{-1} \Omega_0 M_0^{-1} e_1 + o((n|H|)^{-1}).$$

710 Since $\gamma_{x^\star}(z) = \mathbb{E}[W(x^\star, X) | Z = z]$, this is exactly the displayed formula with

$$\Phi(x^\star) = \frac{\mathbb{E}[W(x^\star, X)^2 (Y - m_W(Z; x^\star))^2 | Z = z]}{\mathbb{E}[W(x^\star, X) | Z = z]^2}.$$

711 Under the model $Y = m_C(Z) + \varepsilon$, conditioning on $Z = z$ gives

$$\xi_{x^\star} = (m_C(z) - m_W(z; x^\star)) + \varepsilon,$$

712 so

$$\mathbb{E}[W(x^\star, X)^2 \xi_{x^\star}^2 | Z = z] = \mathbb{E}[W(x^\star, X)^2 \{\sigma^2(z, C) + (m_C(z) - m_W(z; x^\star))^2\} | Z = z],$$

713 because the cross term vanishes after conditioning on (Z, C) and using $\mathbb{E}[\varepsilon | Z, C] = 0$. This
714 gives the stated form of $\Phi(x^\star)$.
715 □

716 Appendix B. Experimental Implementation Details

717 Appendix B.1. Evaluation Protocols

718 Experiment 1 uses the same repeated hold-out design as Experiments 2 and 3: 5 repetitions
719 of an 80%/20% train/test split, with hyperparameters selected by 4-fold cross-validation on each
720 training split and reported metrics aggregated over the held-out test sets. Experiment 2 also fits
721 the local models on $\log(1 + \text{price})$, but model selection and reported metrics are based on raw-
722 price RMSE after inverse transformation of the predictions. Experiment 4 uses a rolling-origin
723 evaluation protocol suited to graph-temporal forecasting: the county-week samples are ordered
724 chronologically, 5 outer train/test windows are formed without shuffling, and hyperparameters
725 are selected by 4-fold time-series cross-validation inside each training window before a single
726 held-out evaluation on the next contiguous time block.

727 *Appendix B.2. Public Data Sources for Experiments 2–4*

728 Experiment 2 uses the archived NYC detailed listings snapshot dated February 13, 2026 from
729 Inside Airbnb ([Inside Airbnb, 2026](#)); the subway context graph is reconstructed from the MTA
730 New York City Transit static GTFS subway feed ([Metropolitan Transportation Authority, 2026](#)).
731 Experiment 3 uses the public `airports.csv` airport-metadata file and `flights-airport.csv`
732 route-count file distributed with Vega’s airport tutorial and datasets repository ([Vega, 2026a,b](#)).
733 Experiment 4 uses the Chickenpox Cases in Hungary benchmark introduced by [Rozemberczki](#)
734 [et al. \(2021\)](#); the implementation reads the public JSON mirror distributed with PyTorch Geo-
735 metric Temporal ([PyTorch Geometric Temporal, 2026](#)).

736 *Appendix B.3. Synthetic Target Construction in Experiment 3*

737 The airport graph is constructed as an undirected graph on direct routes. For the synthetic
738 target, only the PageRank, betweenness, and degree features enter the base signal, after stan-
739 dardization; the latitude and longitude variables are included in the regression features but not
740 in the base-delay formula. The propagation operator uses the traffic-weighted adjacency matrix,
741 row-normalized so each row averages the delays of neighboring airports. The update is iterated
742 for seven steps as displayed in Section 6.3, and the final simulated delay is shifted so that the
743 response remains strictly positive. The graph context kernel used by GC/GRC-LPR is separate
744 from this propagation step: it is computed from *unweighted* shortest-path (hop) distance on the
745 same undirected graph, so the context kernel captures topological locality while the synthetic
746 target still reflects traffic-weighted diffusion.

747 *Appendix B.4. Public Benchmark Construction in Experiment 4*

748 The Hungary chickenpox benchmark is treated as a fixed graph-signal regression problem.
749 The public dataset contains weekly chickenpox counts for 20 counties over 521 weeks on a
750 county graph with 61 undirected edges ([Rozemberczki et al., 2021](#)). For Experiment 4, each
751 node-time pair is flattened into one supervised sample with four lagged weekly counts as predic-
752 tors and the next week’s count as the response, yielding 10,340 samples in total. The raw JSON
753 payload used by the implementation is the public mirror distributed with PyTorch Geometric
754 Temporal ([PyTorch Geometric Temporal, 2026](#)). The graph-aware variants use the same tricube
755 feature kernel as the feature-only models, multiplied by an exponentially decaying hop-distance
756 kernel on the county graph.